Spectroscopic Ellipsometry Analysis
Cap_01242007 Dataset: Ta_2O_5 on Ta Metal

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1 Introduction

This report presents the spectroscopic ellipsometry (SE) analysis of tantalum oxide (Ta_2O_5) thin films deposited on tantalum metal substrates. Six wafers from the Cap_01242007 dataset were analyzed using an optical model developed in collaboration with Neha Singh.

The optical constants for the tantalum metal substrate were obtained using a point-by-point (PBP) fit to the original tantalum metal data (ta_pbp.mat). Barry O'Brien performed the optical constant fit using an aluminum model, which produced sufficiently smooth data for reliable fitting.

2 Optical Model

The layer structure used for fitting is:

Layer	Description
Ambient	Air ($n = 1.0$)
Film	Ta_2O_5 (Cauchy dispersion)
Interface	EMA (Ta_2O_5 + Void)
Substrate	Ta metal (PBP optical constants)

2.1 Cauchy Dispersion for Ta_2O_5

The transparent Ta_2O_5 film is modeled with a Cauchy dispersion:

$$n(\lambda) = A + \frac{B}{\lambda^2} + \frac{C}{\lambda^4} \tag{1}$$

where λ is in micrometers. The extinction coefficient $k = 0$ in the fit range (320–1698 nm).

2.2 EMA Interfacial Layer

An Effective Medium Approximation (EMA) layer between the Ta_2O_5 and Ta metal represents the porous interface created during the oxidation reaction. The EMA mixes Ta_2O_5 optical constants with voids (air, $n = 1.0$).

This interfacial layer is attributed to surface roughness from the columnar grain structure of the tantalum metal. The different mobility of ions during oxidation creates an interface where the stoichiometry differs from the bulk film. Note that substituting surface roughness (srough.mat) for void.mat produces statistically identical results, consistent with this interpretation.

2.3 Fitting Procedure

The fitting was performed using the `refellips` Python package with the Levenberg–Marquardt algorithm. Seven parameters were varied:

- Ta_2O_5 thickness, roughness
- Cauchy coefficients A, B, C
- EMA layer thickness and void fraction

Data was collected at three angles of incidence (65°, 70°, 75°) over the range 246–1698 nm. The fit range was restricted to 320–1698 nm to avoid the UV absorption edge of Ta_2O_5.

3 Results

3.1 Summary

Table 1: Fit results for all six wafers. MSE is the mean squared error in the WVASE convention.

Wafer	Ta_2O_5 (nm)	EMA (nm)	Void (%)	n(600 nm)	MSE
Wafer 1	2158.0	11.99	99.0	2.2019	0.8895
Wafer 2	2157.2	12.17	99.0	2.2064	0.8840
Wafer 3	2163.5	11.51	98.7	2.2009	0.8923
Wafer 4	1806.6	20.00	85.0	2.2228	0.8428
Wafer 5	1818.8	19.03	83.9	2.2122	0.8936
Wafer 6	1804.7	20.00	84.3	2.2247	0.8539

All wafers show excellent fits with MSE below 1.0. Two groups are evident:

- **Wafers 1–3**: Thicker oxide (∼2160 nm), thin EMA (∼12 nm, ∼99% void)
- **Wafers 4–6**: Thinner oxide (∼1810 nm), thicker EMA (∼20 nm, ∼84% void)

The refractive index $n(600\,\text{nm}) \approx 2.20$ is consistent with literature values for amorphous Ta_2O_5.

3.2 Fit Quality

Figures **??**–**??** show the measured (dots) and fitted (lines) Ψ and Δ spectra for each wafer at all three angles of incidence.

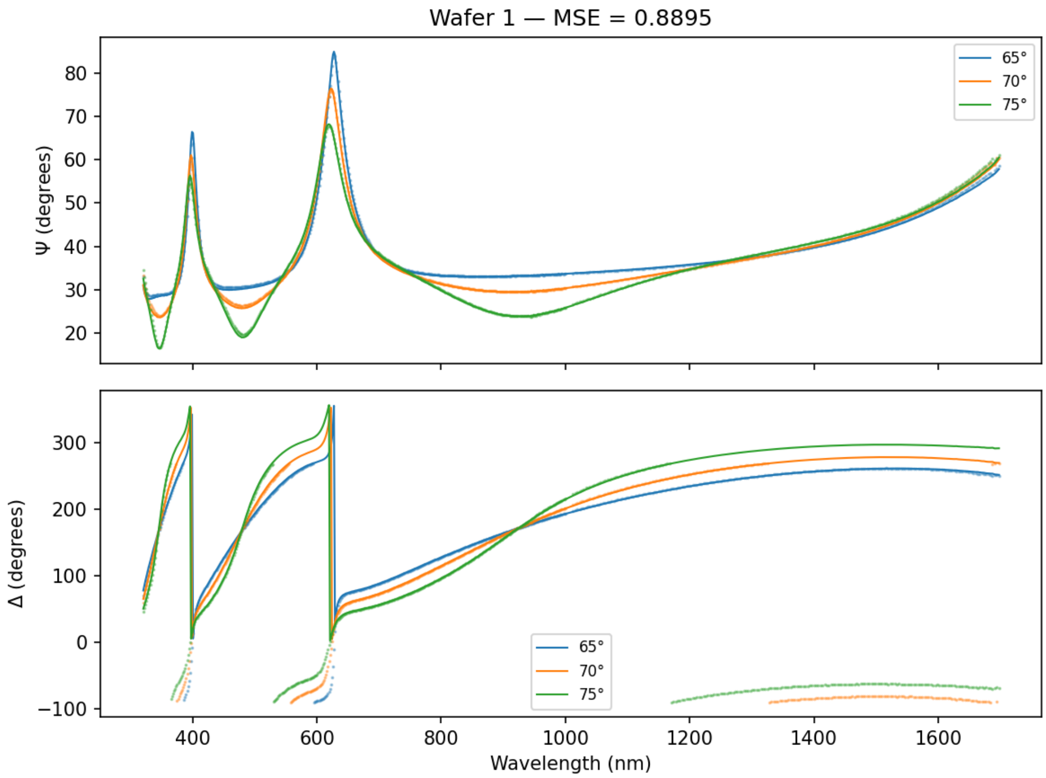

Figure 1: Wafer 1 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

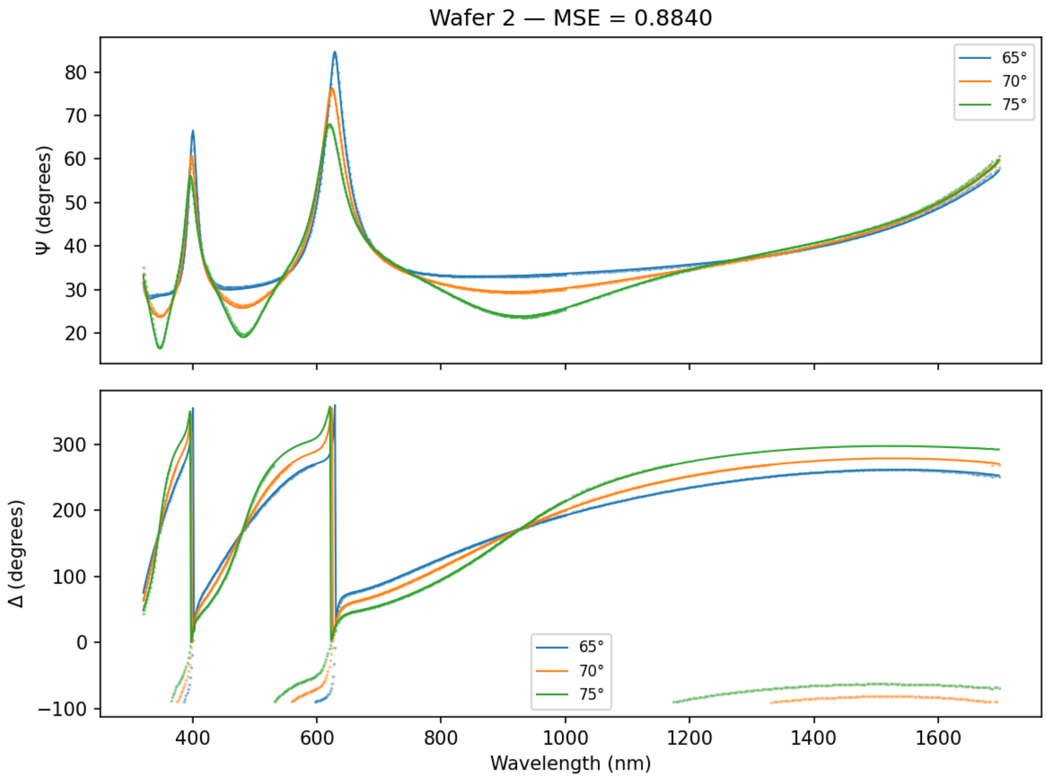

Figure 2: Wafer 2 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

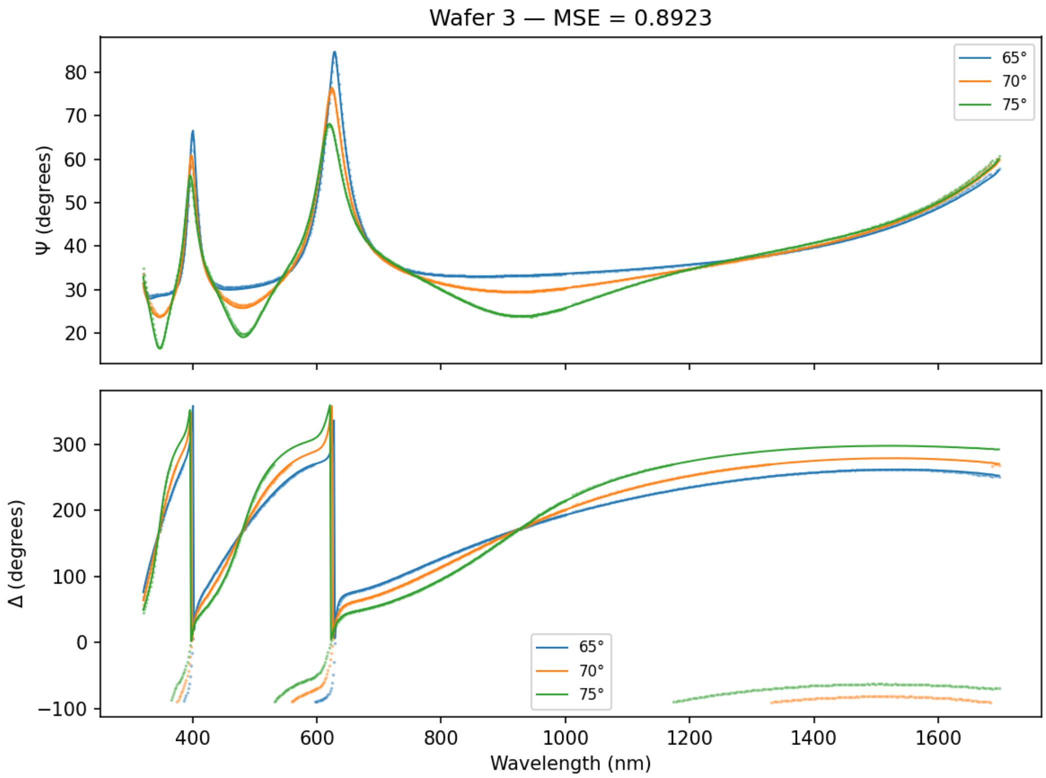

Figure 3: Wafer 3 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

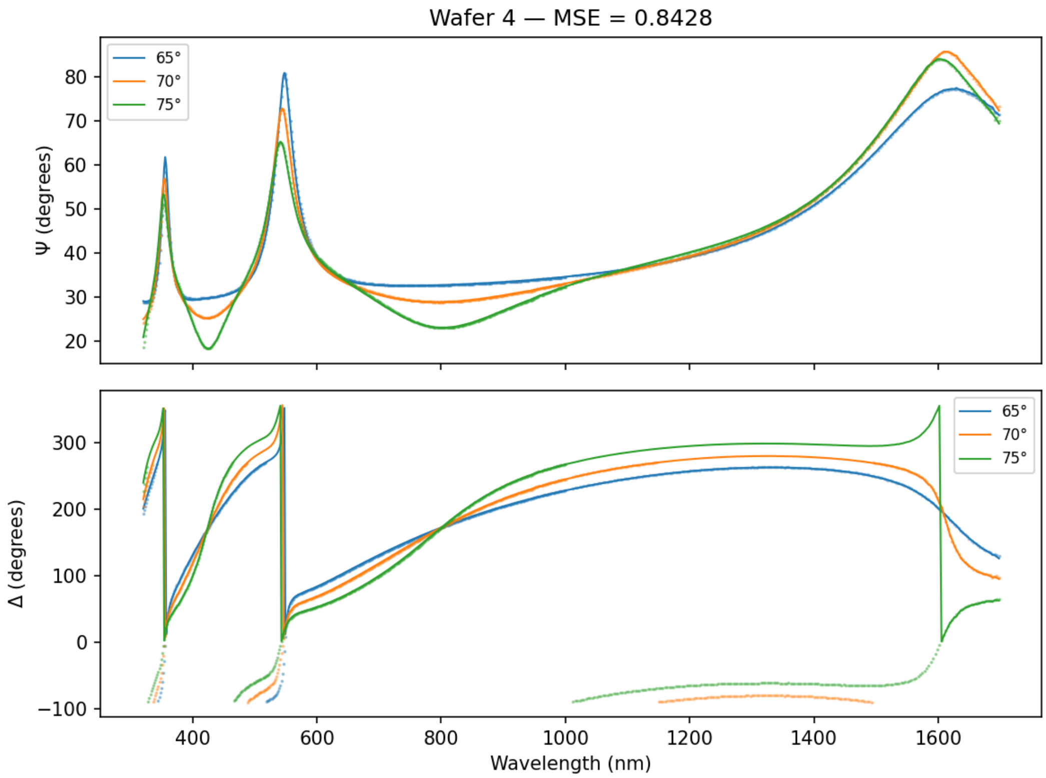

Figure 4: Wafer 4 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

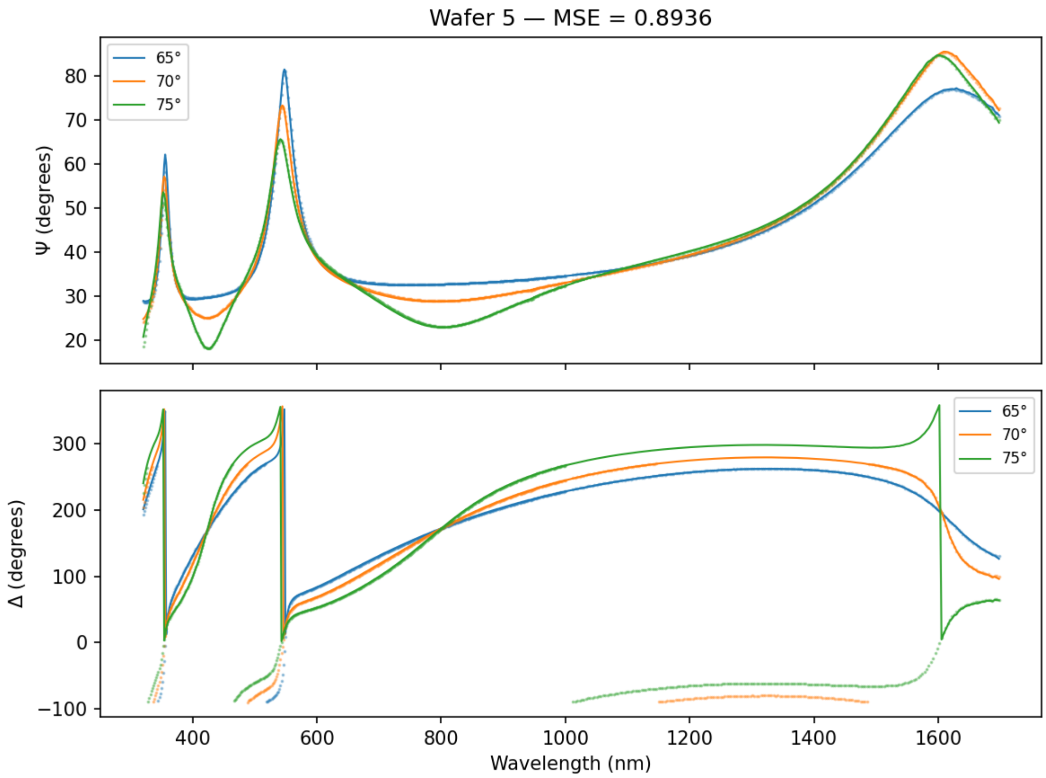

Figure 5: Wafer 5 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

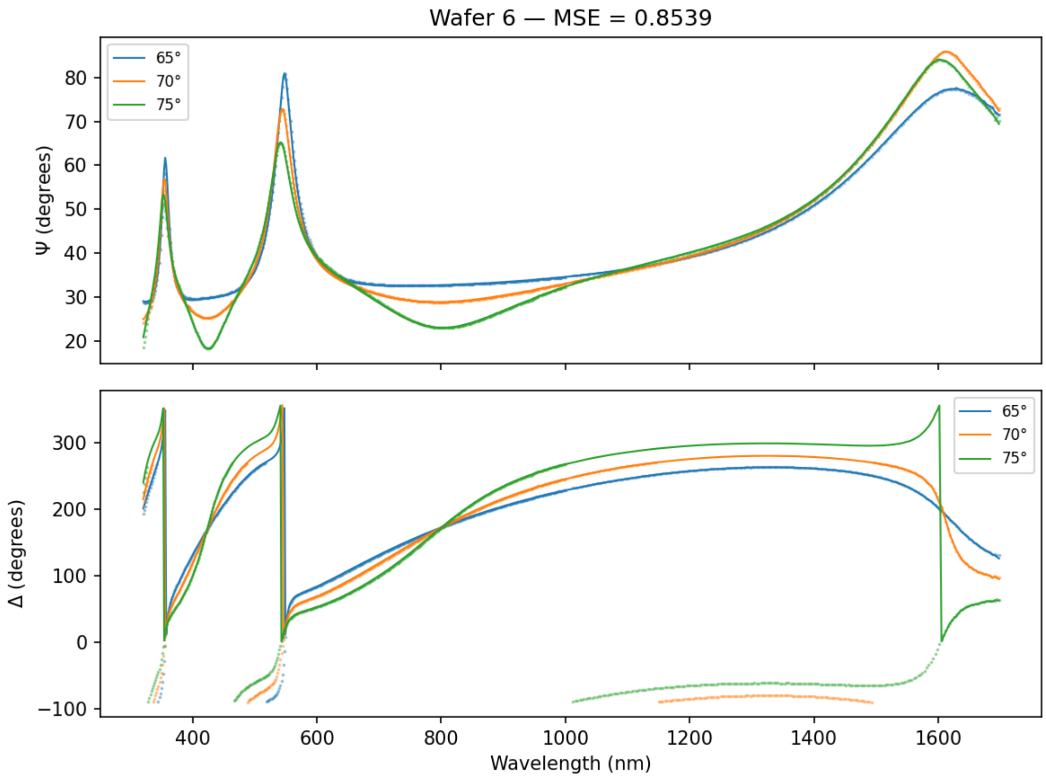

Figure 6: Wafer 6 fit: Ψ and Δ vs. wavelength at 65°, 70°, 75°.

4 Discussion

The two wafer groups (1–3 and 4–6) likely correspond to different process conditions (noted as "high current high voltage" in the data headers). The thicker oxide in wafers 1–3 suggests a longer or more aggressive oxidation, while the higher void fraction in the EMA layer indicates a more porous interface.

The EMA void fraction for wafers 1–3 reaches ∼99%, hitting the upper bound. This suggests the interfacial layer is almost entirely roughness/air rather than a graded composition layer. For wafers 4–6, the ∼84% void fraction is more physically reasonable and aligns with the original finding of $> 80\%$ void.

The EMA layer thickness for wafers 4–6 hits the upper fitting bound (20 nm), suggesting the bound could be relaxed for further investigation. TEM microscopy is recommended to validate the interfacial structure.

5 Conclusion

Neha Singh's model (Cauchy Ta_2O_5 + EMA interfacial layer + Ta metal PBP substrate) provides excellent fits (MSE < 0.9) for all six wafers in the Cap_01242007 dataset. The porous interfacial layer is consistent with the columnar grain roughness mechanism proposed by Curt.